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JEE MAIN · QUESTION PAPER

JEE Main 2022 (Online) 28th July Morning Shift

JEE Main Question Paper with Answer Key

28 Jul 2022	90	20	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

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Chemistry	22 questions
Mathematics	24 questions
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Out of syllabus	20 questions
Answer key	at the end

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Chemistry 22 questions

Q1. MCQ (Single Correct Answer)

Identify the incorrect statement from the following.

(A) A circular path around the nucleus in which an electron moves is proposed as Bohr's orbit.

(B) An orbital is the one electron wave function (ψ) in an atom.

(C) The existence of Bohr's orbits is supported by hydrogen spectrum.

(D) Atomic orbital is characterised by the quantum numbers n and l only.

Open solution

Q2. MCQ (Single Correct Answer)

Which of the following relation is not correct?

(A) $\Delta \mathrm{H}=\Delta \mathrm{U}-\mathrm{P} \Delta \mathrm{V}$

(B) $\Delta \mathrm{U}=\mathrm{q}+\mathrm{W}$

(C) $\Delta \mathrm{S}_{\text{sys}}+\Delta \mathrm{S}_{\text{surr}} \geqslant 0$

(D) $\Delta \mathrm{G}=\Delta \mathrm{H}-\mathrm{T} \Delta \mathrm{S}$

Open solution

Q3. MCQ (Single Correct Answer)

Match List - I with List - II.

	List - I		List - II
(A)	$Cd(s)+2Ni(OH)_3(s) \rightarrow CdO(s)+2Ni(OH)_2(s)+H_2O(l)$	(I)	Primary battery
(B)	$Zn(Hg)+HgO(s) \rightarrow ZnO(s)+Hg(l)$	(II)	Discharging of secondary battery
(C)	$2PbSO_4(s)+2H_2O(l) \rightarrow Pb(s)+PbO_2(s)+2H_2SO_4(aq)$	(III)	Fuel cell
(D)	$2H_2(g)+O_2(g) \rightarrow 2H_2O(l)$	(IV)	Charging of secondary battery

Choose the correct answer from the options given below:

(A) $(\mathrm{A})-(\mathrm{I}),(\mathrm{B})-(\mathrm{II}),(\mathrm{C})-(\mathrm{III}),(\mathrm{D})-(\mathrm{IV})$

(B) $(\mathrm{A})-(\mathrm{IV}),(\mathrm{B})-(\mathrm{I}),(\mathrm{C})-(\mathrm{II}),(\mathrm{D})-(\mathrm{III})$

(C) $(\mathrm{A})-(\mathrm{II}),(\mathrm{B})-(\mathrm{I}),(\mathrm{C})-(\mathrm{IV}),(\mathrm{D})-(\mathrm{III})$

(D) $(\mathrm{A})-(\mathrm{II}),(\mathrm{B})-(\mathrm{I}),(\mathrm{C})-(\mathrm{III}),(\mathrm{D})-(\mathrm{IV})$

Open solution

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Q4. MCQ (Single Correct Answer)

In which of the following pairs, electron gain enthalpies of constituent elements are nearly the same or identical?

(A) Rb and Cs

(B) Na and K

(C) Ar and Kr

(D) I and At

Choose the correct answer from the options given below :

(A) (A) and (B) only **(B)** (B) and (C) only **(C)** (A) and (C) only **(D)** (C) and (D) only

Open solution

Q5. MCQ (Single Correct Answer)

Match List - I with List - II, match the gas evolved during each reaction.

	List - I		List - II
(A)	$(NH_4)_2Cr_2O_7 \xrightarrow{\Delta}$	(I)	H_2
(B)	$KMnO_4 + HCl \rightarrow$	(II)	N_2
(C)	$Al + NaOH + H_2O \rightarrow$	(III)	O_2
(D)	$NaNO_3 \xrightarrow{\Delta}$	(IV)	Cl_2

Choose the correct answer from the options given below :

(A) $(A) - (II), (B) - (III), (C) - (I), (D) - (IV)$

(B) $(A) - (III), (B) - (I), (C) - (IV), (D) - (II)$

(C) $(A) - (II), (B) - (IV), (C) - (I), (D) - (III)$

(D) $(A) - (III), (B) - (IV), (C) - (I), (D) - (II)$

Open solution

Q6. MCQ (Single Correct Answer)

Which of the following has least tendency to liberate H_2 from mineral acids?

(A) Cu **(B)** Mn **(C)** Ni **(D)** Zn

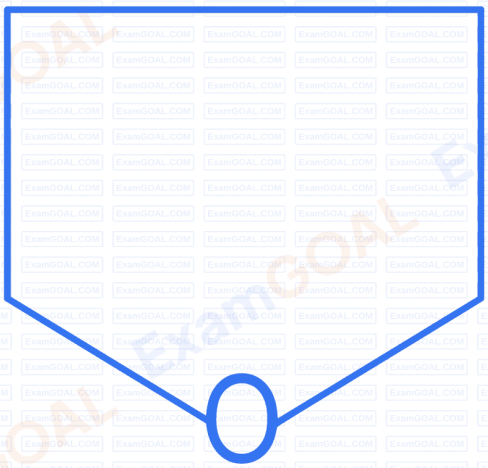
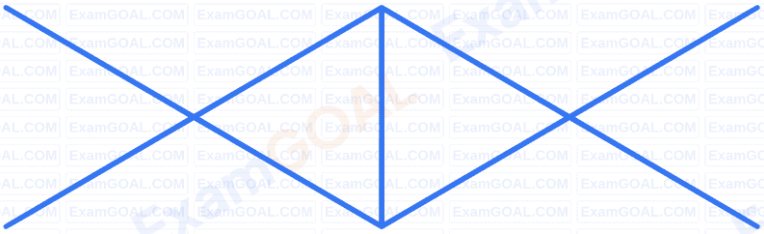
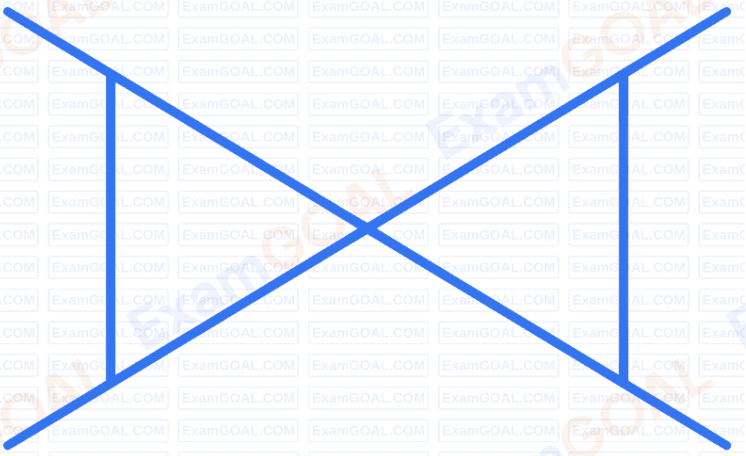
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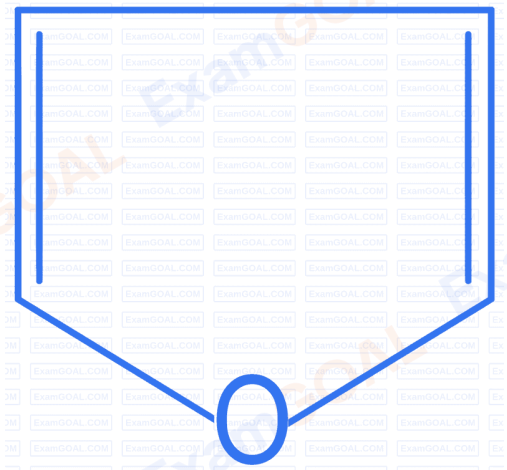
Q7. MCQ (Single Correct Answer)

Open solution

Match List - I with List - II.

	List - I		List - II
(A)		(I)	Spiro compound
(B)		(II)	Aromatic compound
(C)		(III)	Non-planar Heterocyclic compound

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	List - I		List - II
(D)		(IV)	Bicyclo compound

Choose the correct answer from the options given below :

(A) (A) - (II), (B) - (I), (C) - (IV), (D) - (III)

(B) (A) - (IV), (B) - (III), (C) - (I), (D) - (II)

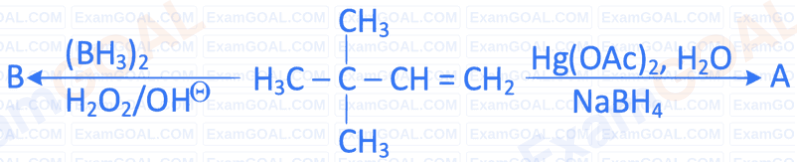
(C) (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

(D) (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

Q8. MCQ (Single Correct Answer)

Open solution

Choose the correct option for the following reactions.



(A) 'A' and 'B' are both Markovnikov addition products.

(B) 'A' is Markovnikov product and 'B' is anti-Markovnikov product.

(C) 'A' and 'B' are both anti-Markovnikov products.

(D) 'B' is Markovnikov and 'A' is anti-Markovnikov product.

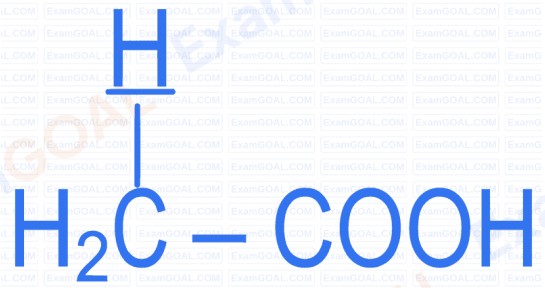
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Q9. MCQ (Single Correct Answer)

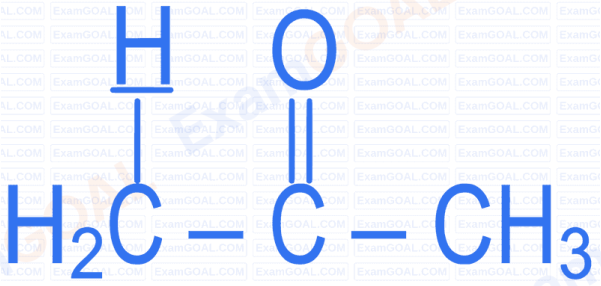
Open solution

Among the following marked proton of which compound shows lowest pK_a value?

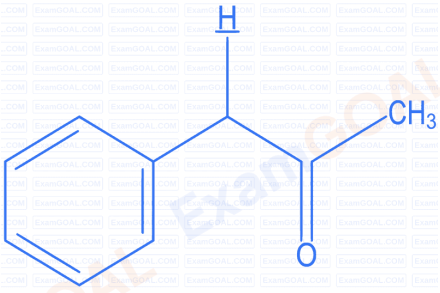
(A)



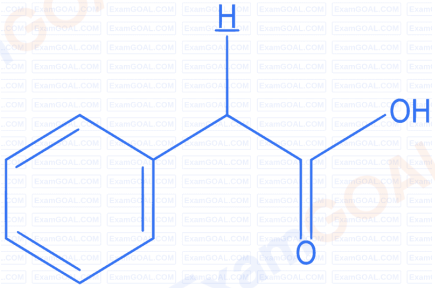
(B)



(C)



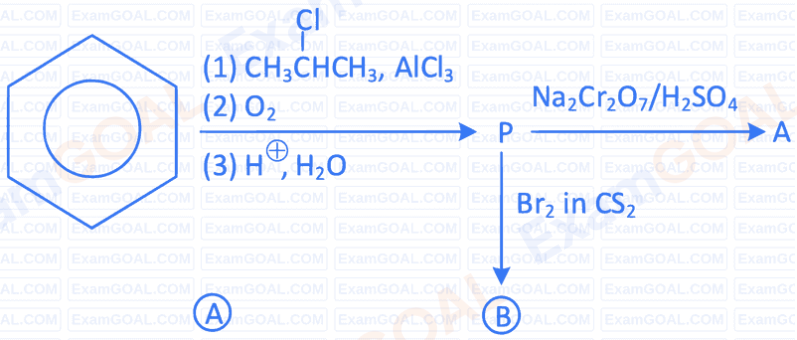
(D)



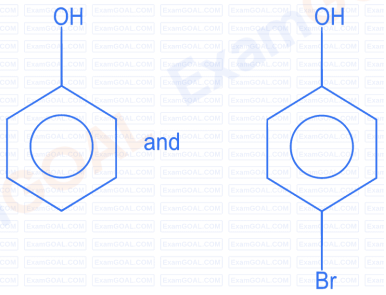
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Q10. MCQ (Single Correct Answer)

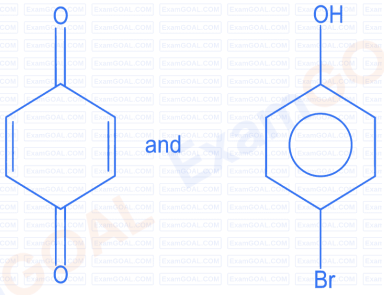
Identify the major products A and B for the below given reaction sequence.



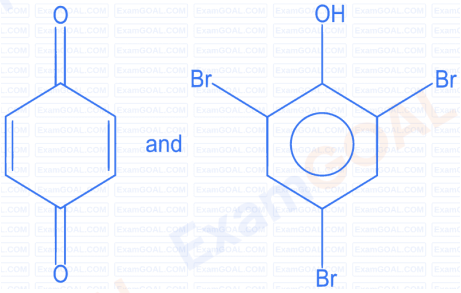
(A)



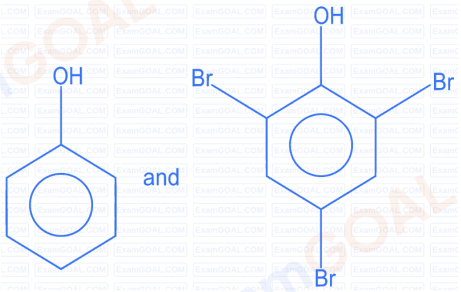
(B)



(C)



(D)



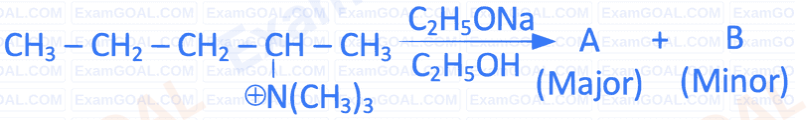
Open solution

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Q11. MCQ (Single Correct Answer)

Open solution

Identify the correct statement for the below given transformation.



(A) $\mathrm{A-CH_3CH_2CH=CH-CH_3}$, $\mathrm{B-CH_3CH_2CH_2CH=CH_2}$, Saytzeff products

(B) $\mathrm{A-CH_3CH_2CH=CH-CH_3}$, $\mathrm{B-CH_3CH_2CH_2CH=CH_2}$, Hofmann products

(C) $\mathrm{A-CH_3CH_2CH_2CH=CH_2}$, $\mathrm{B-CH_3CH_2CH=CHCH_3}$, Hofmann products

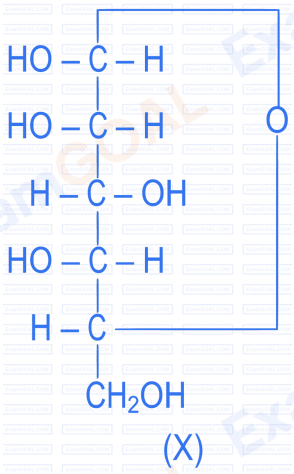
(D) $\mathrm{A-CH_3CH_2CH_2CH=CH_2}$, $\mathrm{B-CH_3CH_2CH=CHCH_3}$, Saytzeff products

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Q12. MCQ (Single Correct Answer)

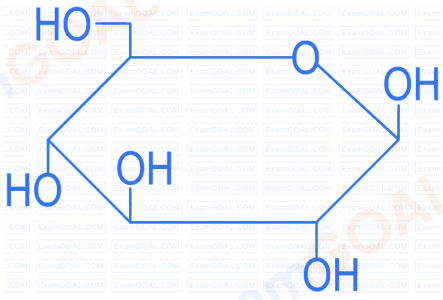
Open solution

For the below given cyclic hemiacetal (X), the correct pyranose structure is :

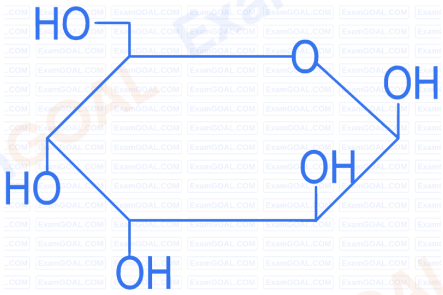


(X)

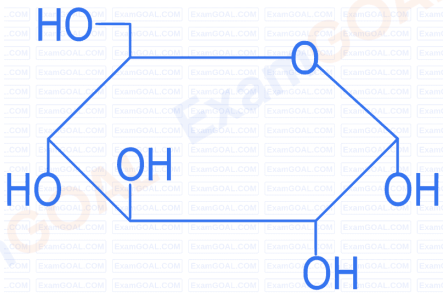
(A)



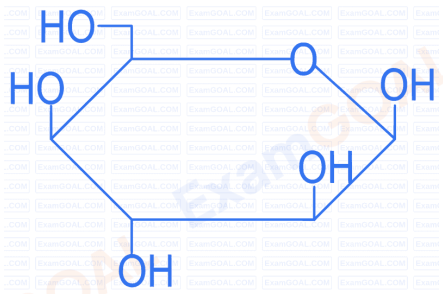
(B)



(C)



(D)



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Q13. MCQ (Single Correct Answer)

Open solution

For kinetic study of the reaction of iodide ion with H_2O_2 at room temperature :

(A) Always use freshly prepared starch solution.

(B) Always keep the concentration of sodium thiosulphate solution less than that of KI solution.

(C) Record the time immediately after the appearance of blue colour.

(D) Record the time immediately before the appearance of blue colour.

(E) Always keep the concentration of sodium thiosulphate solution more than that of KI solution.

Choose the correct answer from the options given below :

(A) $(A), (B), (C)$ only

(B) $(A), (D), (E)$ only

(C) $(D), (E)$ only

(D) $(A), (B), (E)$ only

Q14. Numerical

Open solution

In the given reaction,

$X + Y + 3Z \leftrightarrows XYZ_3$

if one mole of each of X and Y with 0.05 mol of Z gives compound XYZ_3. (Given : Atomic masses of X, Y and Z are 10, 20 and 30 amu, respectively.) The yield of XYZ_3 is _____ g. (Nearest integer)

Q15. Numerical

Open solution

The number of paramagnetic species among the following is _____.

$B_2, Li_2, C_2, C_2^-, O_2^{2-}, O_2^+$ and He_2^+

Q16. Numerical

Open solution

150 g of acetic acid was contaminated with 10.2 g ascorbic acid $(C_6H_8O_6)$ to lower down its freezing point by $\left(x \times 10^{-1}\right)^{\circ}C$. The value of x is _____. (Nearest integer)

[Given $K_f = 3.9$ K kg mol^{-1}; molar mass of ascorbic acid $= 176$ g mol^{-1}]

Q17. Numerical

Open solution

K_a for butyric acid (C_3H_7COOH) is 2×10^{-5}. The pH of 0.2 M solution of butyric acid is _____ $\times 10^{-1}$. (Nearest integer)

[Given $\log 2 = 0.30$]

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Q18. Numerical

For the given first order reaction

$\mathrm{A} \rightarrow \mathrm{B}$

the half life of the reaction is $0.3010\ \mathrm{min}$. The ratio of the initial concentration of reactant to the concentration of reactant at time $2.0\ \mathrm{min}$ will be equal to _____. (Nearest integer)

Open solution

Q19. Numerical

The number of interhalogens from the following having square pyramidal structure is :

$\mathrm{ClF_3}, \mathrm{IF_7}, \mathrm{BrF_5}, \mathrm{BrF_3}, \mathrm{I_2Cl_6}, \mathrm{IF_5}, \mathrm{ClF}, \mathrm{ClF_5}$

Open solution

Q20. Numerical

The disproportionation of $\mathrm{MnO_4^{2-}}$ in acidic medium resulted in the formation of two manganese compounds A and B. If the oxidation state of Mn in B is smaller than that of A, then the spin-only magnetic moment (μ) value of B in BM is _____. (Nearest integer)

Open solution

Q21. Numerical

Total number of relatively more stable isomer(s) possible for octahedral complex $[\mathrm{Cu(en)_2(SCN)_2}]$ will be _____.

Open solution

Q22. Numerical

On complete combustion of $0.492\ \mathrm{g}$ of an organic compound containing C, H and O, $0.7938\ \mathrm{g}$ of $\mathrm{CO_2}$ and $0.4428\ \mathrm{g}$ of $\mathrm{H_2O}$ was produced. The % composition of oxygen in the compound is _____.

Open solution

Mathematics 24 questions

Q23. MCQ (Single Correct Answer)

Let the solution curve of the differential equation $x\,\mathrm{d}y = \left(\sqrt{x^2+y^2}+y\right)\mathrm{d}x, x > 0$, intersect the line $x = 1$ at $y = 0$ and the line $x = 2$ at $y = \alpha$. Then the value of α is :

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$ (C) $-\frac{3}{2}$ (D) $\frac{5}{2}$

Open solution

Q24. MCQ (Single Correct Answer)

Considering only the principal values of the inverse trigonometric functions, the domain of the function $f(x) = \cos^{-1}\left(\frac{x^2-4x+2}{x^2+3}\right)$ is :

(A) $\left(-\infty, \frac{1}{4}\right]$ (B) $\left[-\frac{1}{4}, \infty\right)$

(C) $(-1/3, \infty)$ (D) $\left(-\infty, \frac{1}{3}\right]$

Open solution

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Q25. MCQ (Single Correct Answer)

Let the vectors $\vec{a} = (1+t)\hat{i} + (1-t)\hat{j} + \hat{k}, \vec{b} = (1-t)\hat{i} + (1+t)\hat{j} + 2\hat{k}$ and $\vec{c} = t\hat{i} - t\hat{j} + \hat{k}, t \in \mathbf{R}$ be such that for $\alpha, \beta, \gamma \in \mathbf{R}, \alpha\vec{a} + \beta\vec{b} + \gamma\vec{c} = \vec{0} \Rightarrow \alpha = \beta = \gamma = 0$. Then, the set of all values of t is :

(A) a non-empty finite set
(B) equal to $\mathbf{N}$
(C) equal to $\mathbf{R} - \{0\}$
(D) equal to $\mathbf{R}$

Open solution

Q26. MCQ (Single Correct Answer)

Considering the principal values of the inverse trigonometric functions, the sum of all the solutions of the equation $\cos^{-1}(x) - 2\sin^{-1}(x) = \cos^{-1}(2x)$ is equal to :

(A) 0
(B) 1
(C) $\frac{1}{2}$
(D) $-\frac{1}{2}$

Open solution

Q27. MCQ (Single Correct Answer)

Let a vector $\vec{a}$ has magnitude 9. Let a vector $\vec{b}$ be such that for every $(x, y) \in \mathbf{R} \times \mathbf{R} - \{(0, 0)\}$, the vector $(x\vec{a} + y\vec{b})$ is perpendicular to the vector $(6y\vec{a} - 18x\vec{b})$. Then the value of $|\vec{a} \times \vec{b}|$ is equal to :

(A) $9\sqrt{3}$
(B) $27\sqrt{3}$
(C) 9
(D) 81

Open solution

Q28. MCQ (Single Correct Answer)

For $\mathrm{t} \in (0, 2\pi)$, if ABC is an equilateral triangle with vertices $\mathrm{A}(\sin t, -\cos \mathrm{t}), \mathrm{B}(\cos t, \sin t)$ and $C(a, b)$ such that its orthocentre lies on a circle with centre $\left(1, \frac{1}{3}\right)$, then $\left(a^2 - b^2\right)$ is equal to :

(A) $\frac{8}{3}$
(B) 8
(C) $\frac{77}{9}$
(D) $\frac{80}{9}$

Open solution

Q29. MCQ (Single Correct Answer)

For $\alpha \in \mathbf{N}$, consider a relation R on $\mathbf{N}$ given by $\mathrm{R} = \{(x, y) : 3x + \alpha y$ is a multiple of $7\}$. The relation R is an equivalence relation if and only if :

(A) $\alpha = 14$
(B) α is a multiple of 4
(C) 4 is the remainder when α is divided by 10
(D) 4 is the remainder when α is divided by 7

Open solution

Q30. MCQ (Single Correct Answer)

Out of 60% female and 40% male candidates appearing in an exam, 60% candidates qualify it. The number of females qualifying the exam is twice the number of males qualifying it. A candidate is randomly chosen from the qualified candidates. The probability, that the chosen candidate is a female, is :

(A) $\frac{2}{3}$
(B) $\frac{11}{16}$
(C) $\frac{23}{32}$
(D) $\frac{13}{16}$

Open solution

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Q31. MCQ (Single Correct Answer)

If $y = y(x), x \in (0, \pi/2)$ be the solution curve of the differential equation

$$(\sin^2 2x)\frac{dy}{dx} + (8\sin^2 2x + 2\sin 4x)y = 2e^{-4x}(2\sin 2x + \cos 2x),$$

with $y(\pi/4) = e^{-\pi}$, then $y(\pi/6)$ is equal to :

(A) $\frac{2}{\sqrt{3}}e^{-2\pi/3}$ (B) $\frac{2}{\sqrt{3}}e^{2\pi/3}$

(C) $\frac{1}{\sqrt{3}}e^{-2\pi/3}$ (D) $\frac{1}{\sqrt{3}}e^{2\pi/3}$

Open solution

Q32. MCQ (Single Correct Answer)

Let C be the centre of the circle $x^2 + y^2 - x + 2y = \frac{11}{4}$ and P be a point on the circle. A line passes through the point C, makes an angle of $\frac{\pi}{4}$ with the line CP and intersects the circle at the points Q and R. Then the area of the triangle PQR (in unit ^{2}) is :

(A) 2 (B) $2\sqrt{2}$

(C) $8\sin\left(\frac{\pi}{8}\right)$ (D) $8\cos\left(\frac{\pi}{8}\right)$

Open solution

Q33. MCQ (Single Correct Answer)

The remainder when $7^{2022} + 3^{2022}$ is divided by 5 is :

(A) 0 (B) 2 (C) 3 (D) 4

Open solution

Q34. MCQ (Single Correct Answer)

Let $S_1 = \{z_1 \in \mathbf{C} : |z_1 - 3| = \frac{1}{2}\}$ and $S_2 = \{z_2 \in \mathbf{C} : |z_2 - |z_2 + 1|| = |z_2 + |z_2 - 1||\}$. Then, for $z_1 \in S_1$ and $z_2 \in S_2$, the least value of $|z_2 - z_1|$ is :

(A) 0 (B) $\frac{1}{2}$ (C) $\frac{3}{2}$ (D) $\frac{5}{2}$

Open solution

Q35. MCQ (Single Correct Answer)

If the minimum value of $f(x) = \frac{5x^2}{2} + \frac{\alpha}{x^5}, x > 0$, is 14 , then the value of α is equal to :

(A) 32 (B) 64 (C) 128 (D) 256

Open solution

Q36. MCQ (Single Correct Answer)

Let α, β and γ be three positive real numbers. Let $f(x) = \alpha x^5 + \beta x^3 + \gamma x, x \in \mathbf{R}$ and $g : \mathbf{R} \to \mathbf{R}$ be such that $g(f(x)) = x$ for all $x \in \mathbf{R}$. If $a_1, a_2, a_3, \ldots, a_n$ be in arithmetic progression with mean zero, then the value of $f\left(g\left(\frac{1}{n}\sum_{i=1}^{n} f(a_i)\right)\right)$ is equal to :

(A) 0 (B) 3 (C) 9 (D) 27

Open solution

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Q37. MCQ (Single Correct Answer)

The minimum value of the twice differentiable function $f(x)=\int_0^x e^{x-t}f'(t)dt-(x^2-x+1)e^x$, $x\in\mathbf{R}$, is :

(A) $-\frac{2}{\sqrt{e}}$ (B) $-2\sqrt{e}$ (C) $-\sqrt{e}$ (D) $\frac{2}{\sqrt{e}}$

Open solution

Q38. Numerical

Let S be the set of all passwords which are six to eight characters long, where each character is either an alphabet from $\{A,B,C,D,E\}$ or a number from $\{1,2,3,4,5\}$ with the repetition of characters allowed. If the number of passwords in S whose at least one character is a number from $\{1,2,3,4,5\}$ is $\alpha\times5^6$, then α is equal to _____.

Open solution

Q39. Numerical

Let $P(-2,-1,1)$ and $Q\left(\frac{56}{17},\frac{43}{17},\frac{111}{17}\right)$ be the vertices of the rhombus PRQS. If the direction ratios of the diagonal RS are $\alpha,-1,\beta$, where both α and β are integers of minimum absolute values, then $\alpha^2+\beta^2$ is equal to _____.

Open solution

Q40. Numerical

Let $f:[0,1]\to\mathbf{R}$ be a twice differentiable function in $(0,1)$ such that $f(0)=3$ and $f(1)=5$. If the line $y=2x+3$ intersects the graph of f at only two distinct points in $(0,1)$, then the least number of points $x\in(0,1)$, at which $f''(x)=0$, is _____.

Open solution

Q41. Numerical

If $\int_0^{\sqrt{3}}\frac{15x^3}{\sqrt{1+x^2+\sqrt{(1+x^2)^3}}}dx=\alpha\sqrt{2}+\beta\sqrt{3}$, where α,β are integers, then $\alpha+\beta$ is equal to _____.

Open solution

Q42. Numerical

Let $A=\begin{bmatrix}1&-1\\2&\alpha\end{bmatrix}$ and $B=\begin{bmatrix}\beta&1\\1&0\end{bmatrix}$, $\alpha,\beta\in\mathbf{R}$. Let α_1 be the value of α which satisfies $(A+B)^2=A^2+\begin{bmatrix}2&2\\2&2\end{bmatrix}$ and α_2 be the value of α which satisfies $(A+B)^2=B^2$. Then $|\alpha_1-\alpha_2|$ is equal to _____.

Open solution

Q43. Numerical

For $p,q\in\mathbf{R}$, consider the real valued function $f(x)=(x-p)^2-q$, $x\in\mathbf{R}$ and $q>0$. Let a_1,a_2,a_3 and a_4 be in an arithmetic progression with mean p and positive common difference. If $|f(a_i)|=500$ for all $i=1,2,3,4$, then the absolute difference between the roots of $f(x)=0$ is _____.

Open solution

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Q44. Numerical

Let $x_1, x_2, x_3, \ldots, x_{20}$ be in geometric progression with $x_1 = 3$ and the common ratio $\frac{1}{2}$. A new data is constructed replacing each x_i by $(x_i - i)^2$. If $\bar{x}$ is the mean of new data, then the greatest integer less than or equal to $\bar{x}$ is _____.

Open solution

Q45. Numerical

$\lim\limits_{x \to 0} \left(\frac{(x+2\cos x)^3 + 2(x+2\cos x)^2 + 3\sin(x+2\cos x)}{(x+2)^3 + 2(x+2)^2 + 3\sin(x+2)} \right)^{\frac{100}{x}}$ is equal to _____.

Open solution

Q46. Numerical

The sum of all real values of x for which $\frac{3x^2-9x+17}{x^2+3x+10} = \frac{5x^2-7x+19}{3x^2+5x+12}$ is equal to _____.

Open solution

Physics 24 questions

Q47. MCQ (Single Correct Answer)

The dimensions of $\left(\frac{B^2}{\mu_0}\right)$ will be :

(if μ_0 : permeability of free space and B : magnetic field)

(A) $[ML^2T^{-2}]$

(B) $[MLT^{-2}]$

(C) $[ML^{-1}T^{-2}]$

(D) $[ML^2T^{-2}A^{-1}]$

Open solution

Q48. MCQ (Single Correct Answer)

An NCC parade is going at a uniform speed of $9\ \text{km/h}$ under a mango tree on which a monkey is sitting at a height of $19.6\ \text{m}$. At any particular instant, the monkey drops a mango. A cadet will receive the mango whose distance from the tree at time of drop is: (Given $g = 9.8\ \text{m/s}^2$)

(A) 5 m (B) 10 m (C) 19.8 m (D) 24.5 m

Open solution

Q49. MCQ (Single Correct Answer)

In two different experiments, an object of mass $5\ \text{kg}$ moving with a speed of $25\ \text{ms}^{-1}$ hits two different walls and comes to rest within (i) 3 second, (ii) 5 seconds, respectively. Choose the correct option out of the following :

(A) Impulse and average force acting on the object will be same for both the cases.

(B) Impulse will be same for both the cases but the average force will be different.

(C) Average force will be same for both the cases but the impulse will be different.

(D) Average force and impulse will be different for both the cases.

Open solution

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Q50. MCQ (Single Correct Answer)

A balloon has mass of $10\ \mathrm{g}$ in air. The air escapes from the balloon at a uniform rate with velocity $4.5\ \mathrm{cm/s}$. If the balloon shrinks in $5\ \mathrm{s}$ completely. Then, the average force acting on that balloon will be (in dyne).

Open solution

(A) 3 (B) 9 (C) 12 (D) 18

Q51. MCQ (Single Correct Answer)

If the radius of earth shrinks by 2% while its mass remains same. The acceleration due to gravity on the earth's surface will approximately :

Open solution

(A) decrease by 2% (B) decrease by 4% (C) increase by 2% (D) increase by 4%

Q52. MCQ (Single Correct Answer)

The force required to stretch a wire of cross-section $1\ \mathrm{cm}^2$ to double its length will be : (Given Young's modulus of the wire $= 2 \times 10^{11}\ \mathrm{N/m^2}$)

Open solution

(A) $1 \times 10^7\ \mathrm{N}$ (B) $1.5 \times 10^7\ \mathrm{N}$ (C) $2 \times 10^7\ \mathrm{N}$ (D) $2.5 \times 10^7\ \mathrm{N}$

Q53. MCQ (Single Correct Answer)

Given below are two statements :

Statement I : The average momentum of a molecule in a sample of an ideal gas depends on temperature.

Statement II : The rms speed of oxygen molecules in a gas is v. If the temperature is doubled and the oxygen molecules dissociate into oxygen atoms, the rms speed will become $2v$.

In the light of the above statements, choose the correct answer from the options given below :

Open solution

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q54. MCQ (Single Correct Answer)

In the wave equation

$y = 0.5 \sin \frac{2\pi}{\lambda}(400\mathrm{t} - x)\ \mathrm{m}$

the velocity of the wave will be:

Open solution

(A) 200 m/s (B) $200\sqrt{2}$ m/s (C) 400 m/s (D) $400\sqrt{2}$ m/s

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Q55. MCQ (Single Correct Answer)

Two capacitors, each having capacitance $40\ \mu\mathrm{F}$ are connected in series. The space between one of the capacitors is filled with dielectric material of dielectric constant K such that the equivalence capacitance of the system became $24\ \mu\mathrm{F}$. The value of K will be :

Open solution

(A) 1.5 (B) 2.5 (C) 1.2 (D) 3

Q56. MCQ (Single Correct Answer)

A wire of resistance R_1 is drawn out so that its length is increased by twice of its original length. The ratio of new resistance to original resistance is :

Open solution

(A) 9 : 1 (B) 1 : 9 (C) 4 : 1 (D) 3 : 1

Q57. MCQ (Single Correct Answer)

The current sensitivity of a galvanometer can be increased by :

(A) decreasing the number of turns

(B) increasing the magnetic field

(C) decreasing the area of the coil

(D) decreasing the torsional constant of the spring

Choose the most appropriate answer from the options given below :

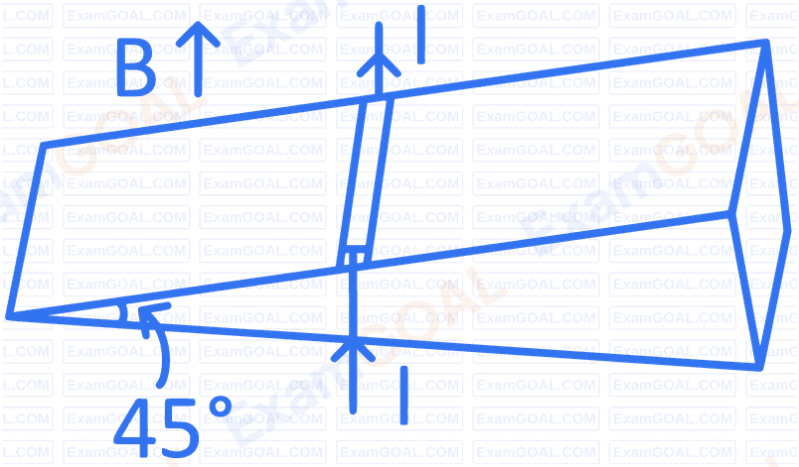
Open solution

(A) (B) and (C) only (B) (C) and (D) only (C) (A) and (C) only (D) (B) and (D) only

Q58. MCQ (Single Correct Answer)

As shown in the figure, a metallic rod of linear density $0.45\ \mathrm{kg\ m^{-1}}$ is lying horizontally on a smooth inclined plane which makes an angle of 45° with the horizontal. The minimum current flowing in the rod required to keep it stationary, when $0.15\ \mathrm{T}$ magnetic field is acting on it in the vertical upward direction, will be :

{Use $g = 10\ \mathrm{m/s^2}$}



Open solution

(A) 30 A (B) 15 A (C) 10 A (D) 3 A

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Q59. MCQ (Single Correct Answer)

Open solution

The equation of current in a purely inductive circuit is $5\sin(49\,\pi t - 30^\circ)$. If the inductance is $30\,\text{mH}$ then the equation for the voltage across the inductor, will be :

{ Let $\pi = \frac{22}{7}$ }

(A) $1.47\sin(49\pi t - 30^\circ)$

(B) $1.47\sin(49\pi t + 60^\circ)$

(C) $23.1\sin(49\pi t - 30^\circ)$

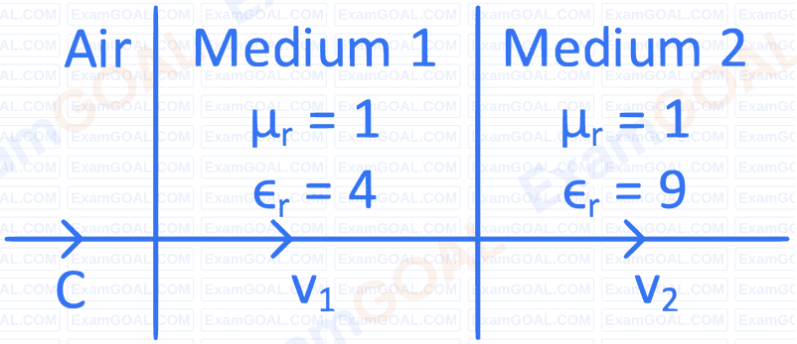
(D) $23.1\sin(49\pi t + 60^\circ)$

Q60. MCQ (Single Correct Answer)

Open solution

As shown in the figure, after passing through the medium 1 . The speed of light v_2 in medium 2 will be :

(Given $c = 3 \times 10^8\ \text{ms}^{-1}$)



(A) $1.0 \times 10^8\ \text{ms}^{-1}$

(B) $0.5 \times 10^8\ \text{ms}^{-1}$

(C) $1.5 \times 10^8\ \text{ms}^{-1}$

(D) $3.0 \times 10^8\ \text{ms}^{-1}$

Q61. MCQ (Single Correct Answer)

Open solution

In normal adjustment, for a refracting telescope, the distance between objective and eye piece is $30\ \text{cm}$. The focal length of the objective, when the angular magnification of the telescope is 2 , will be :

(A) 20 cm

(B) 30 cm

(C) 10 cm

(D) 15 cm

Q62. MCQ (Single Correct Answer)

Open solution

The equation $\lambda = \frac{1.227}{x}\ \text{nm}$ can be used to find the de-Brogli wavelength of an electron. In this equation x stands for :

Where

m = mass of electron

P = momentum of electron

K = Kinetic energy of electron

V = Accelerating potential in volts for electron

(A) $\sqrt{\text{mK}}$

(B) $\sqrt{\text{P}}$

(C) $\sqrt{\text{K}}$

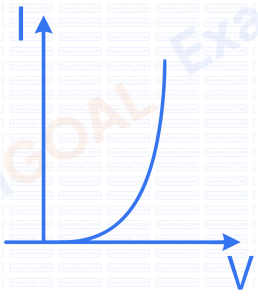
(D) $\sqrt{\text{V}}$

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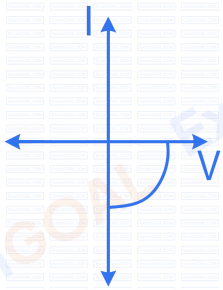
Q63. MCQ (Single Correct Answer)

Identify the solar cell characteristics from the following options :

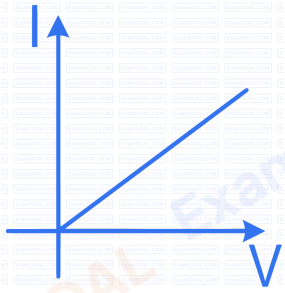
(A)



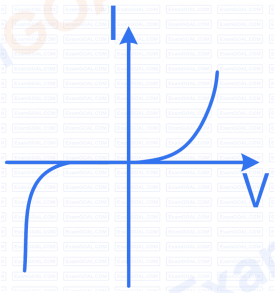
(B)



(C)



(D)



Open solution

Q64. Numerical

If the projection of $2\hat{i}+4\hat{j}-2\hat{k}$ on $\hat{i}+2\hat{j}+\alpha\hat{k}$ is zero. Then, the value of α will be _____.

Open solution

Q65. Numerical

In a Young's double slit experiment, a laser light of 560 nm produces an interference pattern with consecutive bright fringes' separation of 7.2 mm. Now another light is used to produce an interference pattern with consecutive bright fringes' separation of 8.1 mm. The wavelength of second light is _____ nm.

Open solution

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Q66. Numerical

The frequencies at which the current amplitude in an LCR series circuit becomes $\frac{1}{\sqrt{2}}$ times its maximum value, are $212\,\text{rad}\,\text{s}^{-1}$ and $232\,\text{rad}\,\text{s}^{-1}$. The value of resistance in the circuit is $R = 5\,\Omega$. The self inductance in the circuit is _____ mH.

Open solution

Q67. Numerical

Two electric dipoles of dipole moments $1.2 \times 10^{-30}\,\text{Cm}$ and $2.4 \times 10^{-30}\,\text{Cm}$ are placed in two different uniform electric fields of strengths $5 \times 10^4\,\text{NC}^{-1}$ and $15 \times 10^4\,\text{NC}^{-1}$ respectively. The ratio of maximum torque experienced by the electric dipoles will be $\frac{1}{x}$. The value of x is _____.

Open solution

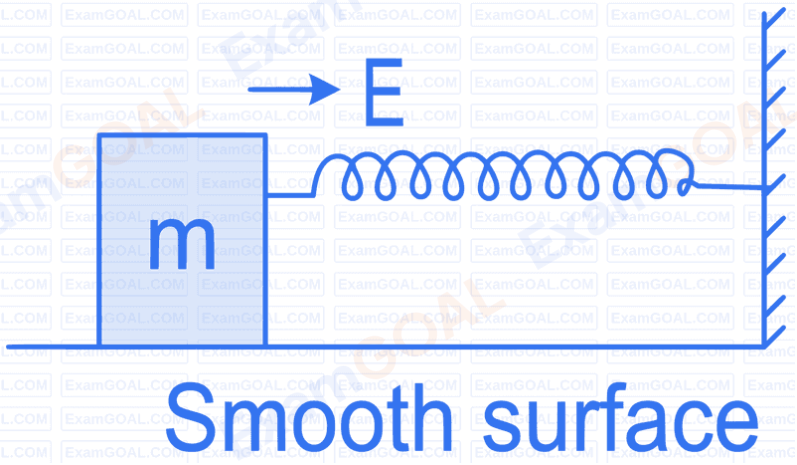
Q68. Numerical

The diameter of an air bubble which was initially $2\,\text{mm}$, rises steadily through a solution of density $1750\,\text{kg}\,\text{m}^{-3}$ at the rate of $0.35\,\text{cms}^{-1}$. The coefficient of viscosity of the solution is _____ poise (in nearest integer). (the density of air is negligible).

Open solution

Q69. Numerical

A block of mass 'm' (as shown in figure) moving with kinetic energy E compresses a spring through a distance $25\,\text{cm}$ when its speed is halved. The value of spring constant of used spring will be $\text{nE}\ \text{Nm}^{-1}$ for $\text{n} =$ _____.



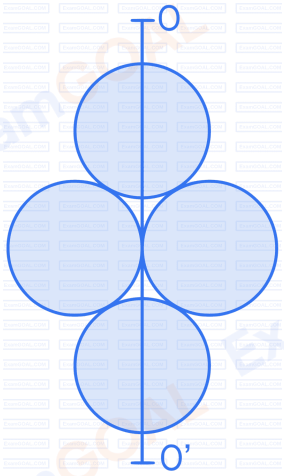
Open solution

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Q70. Numerical

Open solution

Four identical discs each of mass 'M' and diameter 'a' are arranged in a small plane as shown in figure. If the moment of inertia of the system about OO' is $\frac{x}{4}Ma^2$. Then, the value of x will be _____.



Out of syllabus 20 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q71. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Match List - I with List - II.

	List - I Reaction		List - II Catalyst
(A)	$4NH_3(g) + 5O_2(g) \rightarrow 4NO(g) + 6H_2O(g)$	(I)	$NO(g)$
(B)	$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$	(II)	$H_2SO_4(l)$
(C)	$C_{12}H_{22}O_{11}(aq) + H_2O(l) \rightarrow \underset{Glucose}{C_6H_{12}O_6} + \underset{Fructose}{C_6H_{12}O_6}$	(III)	$Pt(s)$
(D)	$2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$	(IV)	$Fe(s)$

Choose the correct answer from the options given below :

(A) $(A) - (II), (B) - (III), (C) - (I), (D) - (IV)$

(B) $(A) - (III), (B) - (II), (C) - (I), (D) - (IV)$

(C) $(A) - (III), (B) - (IV), (C) - (II), (D) - (I)$

(D) (A) - (III), (B) - (II), (C) - (IV), (D) - (I)

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Q72. MCQ (Single Correct Answer) · **Chemistry**

Which of the reaction is suitable for concentrating ore by leaching process ?

(A) $2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$

(B) $Fe_3O_4 + CO \rightarrow 3FeO + CO_2$

(C) $Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2Na[Al(OH)_4]$

(D) $Al_2O_3 + 6Mg \rightarrow 6MgO + 4Al$

Open solution

Q73. MCQ (Single Correct Answer) · **Chemistry**

The metal salts formed during softening of hardwater using Clark's method are :

(A) $Ca(OH)_2$ and $Mg(OH)_2$

(B) $CaCO_3$ and $Mg(OH)_2$

(C) $Ca(OH)_2$ and $MgCO_3$

(D) $CaCO_3$ and $MgCO_3$

Open solution

Q74. MCQ (Single Correct Answer) · **Chemistry**

Which of the following statement is **incorrect**?

(A) Low solubility of LiF in water is due to its small hydration enthalpy.

(B) KO_2 is paramagnetic.

(C) Solution of sodium in liquid ammonia is conducting in nature.

(D) Sodium metal has higher density than potassium metal.

Open solution

Q75. MCQ (Single Correct Answer) · **Chemistry**

Given below are two statements:

Statement I : In polluted water values of both dissolved oxygen and BOD are very low.

Statement II : Eutrophication results in decrease in the amount of dissolved oxygen.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Open solution

Q76. MCQ (Single Correct Answer) · **Chemistry**

Terylene polymer is obtained by condensation of :

(A) Ethane-1,2-diol and Benzene-1,3 dicarboxylic acid

(B) Propane-1, 2-diol and Benzene-1, 4 dicarboxylic acid

(C) Ethane-1,2-diol and Benzene-1, 4 dicarboxylic acid

(D) Ethane-1,2-diol and Benzene-1, 2 dicarboxylic acid

Open solution

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Q77. MCQ (Single Correct Answer) · **Chemistry**

Statements about Enzyme Inhibitor Drugs are given below :

(A) There are Competitive and Non-competitive inhibitor drugs.

(B) These can bind at the active sites and allosteric sites.

(C) Competitive Drugs are allosteric site blocking drugs.

(D) Non-competitive Drugs are active site blocking drugs.

Choose the correct answer from the options given below:

(A) (A), (D) only

(B) (A), (C) only

(C) (A), (B) only

(D) (A), (B), (C) only

Open solution

Q78. Numerical · **Chemistry**

An element M crystallises in a body centred cubic unit cell with a cell edge of $300\ \mathrm{pm}$. The density of the element is $6.0\ \mathrm{g\ cm^{-3}}$. The number of atoms present in $180\ \mathrm{g}$ of the element is _____ $\times 10^{23}$. (Nearest integer)

Open solution

Q79. MCQ (Single Correct Answer) · **Mathematics**

Let the operations $*, \odot \in \{\wedge, \vee\}$. If $(\mathrm{p} * \mathrm{q}) \odot (\mathrm{p} \odot \sim \mathrm{q})$ is a tautology, then the ordered pair $(*, \odot)$ is :

(A) $(\vee, \wedge)$

(B) $(\vee, \vee)$

(C) $(\wedge, \wedge)$

(D) $(\wedge, \vee)$

Open solution

Q80. MCQ (Single Correct Answer) · **Mathematics**

If the tangents drawn at the points P and Q on the parabola $y^2 = 2x - 3$ intersect at the point $R(0, 1)$, then the orthocentre of the triangle PQR is :

(A) (0, 1)

(B) (2, −1)

(C) (6, 3)

(D) (2, 1)

Open solution

Q81. MCQ (Single Correct Answer) · **Mathematics**

Let the matrix $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ and the matrix $B_0 = A^{49} + 2A^{98}$. If $B_n = Adj\,(B_{n-1})$ for all $n \geq 1$, then $\det(B_4)$ is equal to :

(A) 3^{28}

(B) 3^{30}

(C) 3^{32}

(D) 3^{36}

Open solution

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Q82. MCQ (Single Correct Answer) · **Mathematics**

The foot of the perpendicular from a point on the circle $x^2 + y^2 = 1, z = 0$ to the plane $2x + 3y + z = 6$ lies on which one of the following curves?

(A) $(6x + 5y - 12)^2 + 4(3x + 7y - 8)^2 = 1, z = 6 - 2x - 3y$

(B) $(5x + 6y - 12)^2 + 4(3x + 5y - 9)^2 = 1, z = 6 - 2x - 3y$

(C) $(6x + 5y - 14)^2 + 9(3x + 5y - 7)^2 = 1, z = 6 - 2x - 3y$

(D) $(5x + 6y - 14)^2 + 9(3x + 7y - 8)^2 = 1, z = 6 - 2x - 3y$

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

Consider the sequence $a_1, a_2, a_3, \ldots$ such that $a_1 = 1, a_2 = 2$ and $a_{n+2} = \frac{2}{a_{n+1}} + a_n$ for $n = 1, 2, 3, \ldots$. If $\left(\frac{a_1 + \frac{1}{a_2}}{a_3}\right) \cdot \left(\frac{a_2 + \frac{1}{a_3}}{a_4}\right) \cdot \left(\frac{a_3 + \frac{1}{a_4}}{a_5}\right) \ldots \left(\frac{a_{30} + \frac{1}{a_{31}}}{a_{32}}\right) = 2^{\alpha}\left({}^{61}C_{31}\right)$, then α is equal to :

(A) -30 (B) -31 (C) -60 (D) -61

Open solution

Q84. Numerical · **Mathematics**

For the hyperbola $H : x^2 - y^2 = 1$ and the ellipse $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b > 0$, let the

(1) eccentricity of E be reciprocal of the eccentricity of H, and

(2) the line $y = \sqrt{\frac{5}{2}}x + K$ be a common tangent of E and H.

Then $4\left(a^2 + b^2\right)$ is equal to _____.

Open solution

Q85. MCQ (Single Correct Answer) · **Physics**

A Carnot engine has efficiency of 50%. If the temperature of sink is reduced by $40^{\circ}C$, its efficiency increases by 30%. The temperature of the source will be:

(A) 166.7 K (B) 255.1 K (C) 266.7 K (D) 367.7 K

Open solution

Q86. MCQ (Single Correct Answer) · **Physics**

The half life period of a radioactive substance is 60 days. The time taken for $\frac{7}{8}$th of its original mass to disintegrate will be :

(A) 120 days (B) 130 days (C) 180 days (D) 20 days

Open solution

Q87. MCQ (Single Correct Answer) · **Physics**

In the case of amplitude modulation to avoid distortion the modulation index (μ) should be :

(A) $\mu \leq 1$ (B) $\mu \geq 1$ (C) $\mu = 2$ (D) $\mu = 0$

Open solution

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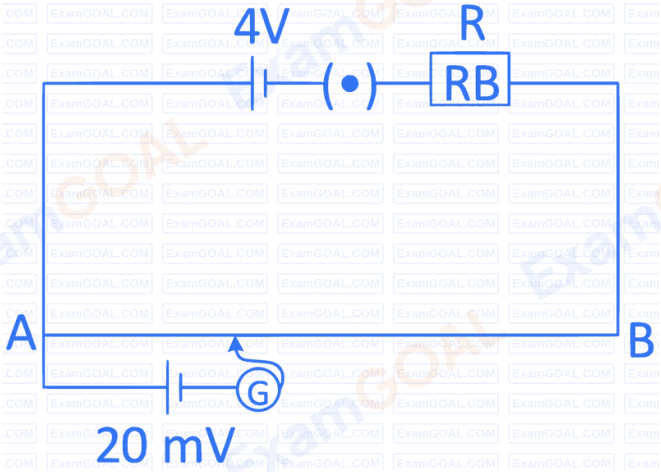
Q88. Numerical · **Physics**

A freshly prepared radioactive source of half life 2 hours 30 minutes emits radiation which is 64 times the permissible safe level. The minimum time, after which it would be possible to work safely with source, will be _____ hours.

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Q89. Numerical · **Physics**

As shown in the figure, a potentiometer wire of resistance $20\,\Omega$ and length $300\,\text{cm}$ is connected with resistance box (R.B.) and a standard cell of emf $4\,\text{V}$. For a resistance 'R' of resistance box introduced into the circuit, the null point for a cell of $20\,\text{mV}$ is found to be $60\,\text{cm}$. The value of 'R' is _____ Ω.



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Q90. Numerical · **Physics**

The frequency of echo will be _____ Hz if the train blowing a whistle of frequency 320 Hz is moving with a velocity of 36 km/h towards a hill from which an echo is heard by the train driver. Velocity of sound in air is 330 m/s.

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Answer Key

Chemistry

1. **D**	2. **A**	3. **C**	4. **C**	5. **C**
6. **A**	7. **C**	8. **B**	9. **C**	10. **B**
11. **C**	12. **D**	13. **A**	14. **2**	15. **4**
16. **15**	17. **27**	18. **100**	19. **3**	20. **4**
21. **3**	22. **46**			

Mathematics

23. **B**	24. **B**	25. **C**	26. **A**	27. **B**
28. **B**	29. **D**	30. **A**	31. **A**	32. **B**
33. **C**	34. **C**	35. **C**	36. **A**	37. **A**
38. **7073**	39. **450**	40. **2**	41. **10**	42. **2**
43. **50**	44. **142**	45. **1**	46. **6**	

Physics

47. **C**	48. **A**	49. **B**	50. **B**	51. **D**
52. **C**	53. **D**	54. **C**	55. **A**	56. **A**
57. **D**	58. **A**	59. **D**	60. **A**	61. **A**
62. **D**	63. **B**	64. **5**	65. **630**	66. **250**
67. **6**	68. **11**	69. **24**	70. **3**	

Out of syllabus

71. **C**	72. **C**	73. **B**	74. **A**	75. **D**
76. **C**	77. **C**	78. **22**	79. **B**	80. **B**
81. **C**	82. **B**	83. **C**	84. **3**	85. **C**
86. **C**	87. **A**	88. **15**	89. **780**	90. **340**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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